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$$T_4(f, 0)(x) \text{ voor } f(x) = \frac{1}{1+x}$$

$$f(x) = (1+x)^{-1} \qquad f(0) = 1$$

$$f'(x) = -1(1+x)^{-2} \qquad f'(0) = -1$$

$$f''(x) = 2(1+x)^{-3} \qquad f''(0) = 2$$

$$f'''(x) = -6(1+x)^{-4} \qquad f'''(0) = -6$$

$$f^{iv}(x) = 24(1+x)^{-5} \qquad f^{iv}(0) = 24$$

$$\begin{aligned} \Rightarrow T_4\left(\frac{1}{1+x}, 0\right)(x) &= \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k \\ &= \frac{1}{0!} x^0 + \frac{-1}{1!} x^1 + \frac{2}{2!} x^2 + \frac{-6}{3!} x^3 + \frac{24}{4!} x^4 \\ &= 1 - x + x^2 - x^3 + x^4 \end{aligned}$$

References